JARID2-Related Neurodevelopmental Disorder — Comprehensive Disease Characterization

Causal gene: *JARID2* (Jumonji And AT-Rich Interaction Domain 2; HGNC:6826; NCBI Gene 3720; UniProt Q92833). MONDO:0859306 · OMIM:620098 (NEURODEVELOPMENTAL DISORDER WITH DELAYED SPEECH AND BEHAVIORAL ABNORMALITIES; gene OMIM:601594) · Category: Mendelian, autosomal dominant.

> Note on identity: The disease name is generic ("neurodevelopmental disorder"), so this report is anchored to the MONDO/OMIM identifiers and the causal gene **JARID2** given in the brief. JARID2 sits on chromosome **6p22.3** and encodes a Polycomb Repressive Complex 2 (PRC2) accessory subunit. This should not be confused with the many other serially numbered neurodevelopmental disorders (e.g., those caused by SETD5, ANKRD11, KMT2A), which are distinct chromatinopathies.

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1. Disease Information

Overview. JARID2-Related Neurodevelopmental Disorder is a rare, recently delineated Mendelian chromatinopathy caused by heterozygous loss-of-function (and, less commonly, gain-of-dosage) alterations of *JARID2*. It is characterized by global developmental delay and/or intellectual/learning disability (ID), speech delay, autism/behavioral-neuropsychiatric abnormalities, hypotonia, and a recognizable facial gestalt (prominent supraorbital ridges, deep-set eyes, dark infraorbital circles, midface hypoplasia); palatal abnormalities and congenital heart defects occur in a subset. Gene–disease validity was formally established by Verberne et al. 2021 in a cohort of 16 individuals from 15 families (P 33077894): *"We report that JARID2 haploinsufficiency leads to a clinically distinct neurodevelopmental syndrome, thus establishing gene-disease validity for the purpose of diagnostic reporting."* It was

independently confirmed and phenotypically expanded by Cadieux-Dion et al. 2022 (
P 35533077
, 7 individuals/5 families).

Key identifiers. - MONDO: **MONDO:0859306** - OMIM (phenotype): **620098**; OMIM (gene JARID2): **601594** - Gene: HGNC:6826 · NCBI Gene:3720 · Ensembl ENSG00000008083 · UniProt Q92833 - Orphanet: JARID2-related disorder is covered under 6p22-p24 deletion / rare chromatinopathies (no unique high-visibility ORPHAcode widely assigned at time of writing — **not definitively recorded**). - ICD-11: closest mapping **LD2F.1Y** / **6A00.Z** style codes for syndromic intellectual developmental disorders (no JARID2-specific code — **not available**). - MeSH: no JARID2-specific descriptor; indexed under *Intellectual Disability* and *Neurodevelopmental Disorders*.

Synonyms / alternative names. "JARID2 haploinsufficiency syndrome"; "JARID2-related intellectual disability"; historically a subset of "6p22-p24 deletion syndrome." Older gene aliases: **JMJ** (jumonji), Jumonji homolog.

Source of information. Entirely **disease-level/individual-patient case-series literature** (aggregated published cohorts and single case reports), not EHR-derived population data. Evidence is predominantly human clinical (case series) plus model-organism and in-vitro mechanistic work.

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2. Etiology

Primary cause — genetic. The disorder is caused by heterozygous germline alterations of *JARID2*: (i) intragenic/whole-gene **deletions** (including contiguous 6p22-p24 deletions encompassing JARID2), and (ii) **single-nucleotide loss-of-function variants** (nonsense, frameshift, splice) predicted to trigger nonsense-mediated mRNA decay → haploinsufficiency (Verberne 2021,
P 33077894
: *"Several of these variants are likely to result in haploinsufficiency due to nonsense-mediated messenger RNA (mRNA) decay."*). A de novo partial **duplication** (exons 1–7) causing increased

expression has also been reported (Viitasalo 2022, P 35979655), indicating bidirectional dosage sensitivity.

Genetic risk factors. The causal variant *is* the risk factor (monogenic, high-penetrance). Most cases are **de novo**. Candidate **modifier** influences: genetic background strongly modulates penetrance in the mouse ortholog (strain-dependent neural-tube defects; Takeuchi 1999, P 10446263), suggesting unidentified human modifier loci. Adjacent 6p22.3 gene **ATXN1** (and DTNBP1, GMPR) contributes to the broader phenotype in large contiguous deletions (Barøy 2013, P 23294540).

Environmental risk factors. None established. No toxin, infection, or lifestyle exposure is implicated. Advanced parental age is a generic contributor to de novo mutation rate but is not specifically documented here.

Protective factors (genetic or environmental). None identified / not applicable.

Gene–environment interactions. Not documented in humans. By analogy with the mouse (variable neural-tube-defect penetrance across strains), genetic background likely interacts with the primary lesion, but no specific GxE interaction is proven.

3. Phenotypes

Frequencies are qualitative/approximate, drawn from small published cohorts (Barøy 2013 n≈5; Verberne 2021 n=16). Onset is **congenital/early childhood**; course is **static (non-progressive) but lifelong**.

Phenotype	Type	Frequency	HPO term
Global developmental delay	clinical sign	~100%	HP:0001263
Intellectual disability (mild–moderate, variable)	clinical sign	high (most)	HP:0001249
Delayed speech/language development	clinical sign	common	HP:0000750
Motor delay / gait disturbance	clinical sign	common	HP:0001270 / HP:0001288
Muscular hypotonia	clinical sign	common	HP:0001252
Behavioral/neuropsychiatric abnormality (incl. autistic features, ADHD)	behavioral	common	HP:0000708 / HP:0000729
Prominent supraorbital ridges	physical	characteristic	HP:0000336
Deeply set eyes	physical	characteristic	HP:0000490
Dark infraorbital circles	physical	characteristic	HP:0000652 (periorbital)
Midface retrusion/hypoplasia	physical	characteristic	HP:0011800
Abnormal facial gestalt	physical	characteristic	HP:0001999
Autism spectrum / autistic behavior	behavioral	subset	HP:0000729
Palatal abnormality (high-arched/cleft)	physical	subset	HP:0000174
Congenital heart defect	physical	subset	HP:0001627

Severity: mild to moderate, **variable** between individuals (variable expressivity — some carrier parents are only mildly affected; Cadieux-Dion 2022,).

P 35533077

Progression: stable/static developmental disorder; not neurodegenerative. **Seizures, structural brain, growth, and organ anomalies** are reported variably in large contiguous

deletions but are **not consistently attributable to JARID2 alone — data incomplete.** Palatal and cardiac anomalies are now recognized as part of the single-gene phenotype in a subset ( ). PMID 35533077

Quality-of-life impact. Lifelong impact on learning, communication, adaptive function, and independence; behavioral comorbidity affects caregiver burden. No disease-specific EQ-5D/SF-36/PROMIS data exist (**not available**); impact inferred from the ID/ASD-spectrum QoL literature.

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4. Genetic / Molecular Information

Causal gene. *JARID2* (HGNC:6826; OMIM 601594), 6p22.3, ~28 exons, encoding a ~1246-aa nuclear protein with a **JmjN, JmjC** (enzymatically dead — lacks key Fe(II)/2-OG-coordinating residues), an **ARID/Bright DNA-binding** domain, and a zinc finger. It is a **catalytically inactive Jumonji-family** protein.

Pathogenic variants. - **Types:** whole-gene & intragenic deletions; CNV at 6p22-p24; nonsense/frameshift/splice LoF SNVs (NMD → haploinsufficiency); at least one partial **duplication** (gain). A homozygous **missense** (c.899G>A, p.Arg300Gln) underlies a separate recessive reproductive phenotype (Sun 2025,). PMID 40886009

- **Classification (ACMG/AMP):** truncating LoF variants → Pathogenic/Likely pathogenic (PVS1-supported); many missense remain **VUS**, where DNA-methylation episignature testing can help reclassification (chromatinopathy EpiSign framework, Koparir 2026,). PMID 41957673

- **Allele frequency:** pathogenic alleles are **absent/ultra-rare** in gnomAD; JARID2 is a predicted **loss-of-function-intolerant** gene (consistent with haploinsufficiency; precise pLI/LOEUF to confirm in gnomAD — qualitative). - **Origin:** predominantly **germline, de novo**. No somatic-disease role in this NDD. - **Functional consequence: loss of function / haploinsufficiency** (dominant); duplication implies **dosage-gain** also pathogenic.

Modifier genes. Within 6p22.3: **ATXN1** (independent ID contributor), DTNBP1, GMPR (Barøy 2013). Genetic background acts as modifier in mouse.

Epigenetic information. JARID2 is itself an epigenetic regulator (PRC2.2). Loss alters genome-wide **H3K27me3** at developmental targets; a **blood DNA-methylation episignature** is being defined for diagnostic use (van der Laan 2023, P 37762546; chromatinopathy EpiSign cohorts, P 41957673).

Chromosomal abnormalities. Interstitial **6p22-p24 deletions** (0.1–17 Mb) are the historical presentation (Barøy 2013; Verberne 2021). DECIPHER/ClinVar/dbVar are relevant repositories.

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5. Environmental Information

- **Environmental factors:** none implicated — **not applicable** (monogenic disorder).
- **Lifestyle factors:** none — **not applicable**.
- **Infectious agents:** none — **not applicable**.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A heterozygous **LoF variant or deletion in *JARID2*** (or, alternatively, a dosage-increasing duplication) **reduces (or raises) functional JARID2 protein** — *demonstrated* (Verberne 2021; Viitasalo 2022).
2. Altered JARID2 dosage **impairs assembly/targeting of the PRC2.2 variant complex** (JARID2–AEBP2–EZH2–SUZ12–EED–RBBP4/7) to CpG-island nucleation sites — *demonstrated in vitro* (Loh & Veenstra 2022, P 35997369; Granat 2026, P 41650228).

3. This **reduces initiation/spreading of the PRC2 "read-write" H3K27me3 loop** (JARID2 normally mimics H3K27me3 to seed EZH2 methyltransferase activity) — *demonstrated* (Agius 2026,

   P 41785858

).
4. Reduced **H3K27me3 at developmental loci (e.g., HOX and lineage-specifying genes) leads to derepression/mis-timing of gene silencing** — *demonstrated in stem-cell/mouse models*; *inferred* for human neurons.
5. Dysregulated Polycomb target expression **disrupts the normal program of neural progenitor proliferation, differentiation, and neuronal maturation** during a critical embryonic/early-postnatal window — *inferred from mouse neurulation phenotype* (Takeuchi 1999).
6. Aberrant neurodevelopment **results in developmental delay, intellectual disability, speech delay, and behavioral abnormalities**, with concurrent craniofacial patterning changes producing the facial gestalt — *clinical correlation.*
7. **Branch A (canonical, PRC2/epigenetic):** steps 2–4 above.
8. **Branch B (non-canonical signaling):** JARID2 dysfunction **attenuates ERK1/2 (MAPK) pathway activation**, shown for the recessive p.Arg300Gln allele in oocytes/zygotes (Sun 2025,

   P 40886009

) — *demonstrated for reproductive phenotype; relevance to NDD inferred.*
9. **Branch C (cardiac, developmental):** in mouse, Jarid2 loss **disrupts outflow-tract morphogenesis and trabecular myocyte proliferation** (Takeuchi 1999) — explains congenital heart involvement seen in some deletion patients.

Detail by category

- **Molecular pathways:** Polycomb/PRC2 H3K27-methylation axis (primary); **MAPK/ERK1/2** signaling (secondary, Sun 2025). Downstream: HOX and developmental transcription-factor networks. Suggested pathway refs: Reactome *PRC2 methylates histones and DNA*; KEGG not specific.
- **Cellular processes:** chromatin-mediated transcriptional repression; cell-fate commitment; neural progenitor **proliferation/differentiation**; (in mouse) cardiomyocyte proliferation. GO: **GO:0006342** chromatin silencing; **GO:0070734** histone H3-K27 methylation; **GO:0021915**

neural tube development; **GO:0030182** neuron differentiation; **GO:0045596** negative regulation of cell differentiation.

- **Protein dysfunction:** haploinsufficiency (reduced scaffold for PRC2 recruitment); missense may act via reduced pathway activation without loss of abundance (Sun 2025). No aggregation. Enzymatically dead JmjC (scaffold, not demethylase).
- **Epigenetic changes:** decreased H3K27me3 at targets; a reproducible **blood DNAm episignature** (diagnostic biomarker).
- **Transcriptomics:** RNA-seq shows dysregulated PRC2 target genes (Viitasalo 2022).
- **Immune/metabolic/tissue-damage categories: not applicable / not reported.**
- **Cell types (CL):** neural stem/progenitor cell **CL:0000047**; glutamatergic/GABAergic neurons **CL:0000679** / **CL:0000617**; cardiomyocyte **CL:0000746** (mouse). JARID2 "is expressed in human neurons" (Verberne 2021).

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7. Anatomical Structures Affected

- **Organ level — primary: brain / central nervous system** (nervous system; UBERON:0000955 brain, UBERON:0001016 nervous system). Secondary/variable: **heart** (congenital defects in some deletion cases; UBERON:0000948), craniofacial skeleton (midface; UBERON:0002104/face).
- **Body systems:** nervous system (primary); cardiovascular and craniofacial (secondary/variable).
- **Tissue/cell level:** nervous tissue; neural progenitors and neurons (CL:0000047, CL:0000540); in mouse also neural plate epithelium (UBERON:0005732) and cardiac trabecular myocytes (CL:0000746).
- **Subcellular level: nucleus / nucleoplasm / chromatin** (GO:0005634 nucleus; GO:0000785 chromatin) — JARID2 is a chromatin-bound nuclear factor.
- **Localization / lateralization:** CNS involvement is **bilateral/global**; craniofacial features are symmetric/midline (midface).

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8. Temporal Development

- **Onset: congenital / early infancy**; developmental concerns typically recognized in infancy–early childhood. Onset pattern **insidious/chronic** (developmental, not acute).
- **Progression: static (non-progressive)** neurodevelopmental disorder; disabilities are **lifelong**. No neurodegeneration or disease "staging." Progression rate: not applicable (stable).
- **Course pattern:** stable/chronic; behavioral features may fluctuate but the underlying disorder does not relapse-remit.
- **Remission:** none spontaneous; developmental gains occur with therapy (treatment-responsive functional improvement, not cure).
- **Critical period:** embryonic–early-postnatal neurodevelopment is the mechanistically vulnerable window (PRC2-dependent fate specification); early intervention (therapy/education) is the clinical window of opportunity.

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9. Inheritance and Population

- **Epidemiology: ultra-rare**; precise prevalence/incidence **not established** (only a few dozen published individuals worldwide as of 2025). No Orphanet point-prevalence figure.
- **Inheritance: autosomal dominant**; most cases **de novo**, but **inheritance from a mildly affected parent is documented** (Verberne 2021 had one inherited case; Cadieux-Dion 2022, P 35533077 reported additional transmitted variants from mildly affected parents). A separate **autosomal recessive** reproductive phenotype arises from biallelic missense (Sun 2025).
- **Penetrance:** high but **not strictly complete** — mildly affected transmitting parents indicate **variable/reduced penetrance** in some families; mouse data show **background-dependent penetrance** for structural defects.
- **Expressivity: highly variable** (severity of ID, autism/behavior, palatal and cardiac involvement) even within families (P 35533077).
- **Genetic anticipation: not applicable** (no repeat expansion).

- **Germline mosaicism:** not documented but theoretically possible for recurrence risk — **not reported**.
- **Founder effects / geographic variant clustering: none known**.
- **Consanguinity:** relevant only to the **recessive** reproductive phenotype.
- **Carrier frequency:** pathogenic alleles ultra-rare in gnomAD (**effectively not a carrier-screening target**).
- **Sex ratio:** no strong skew reported for the dominant NDD (**data limited**); the recessive reproductive/infertility phenotype is female-specific (maternal-effect gene).
- **Age distribution:** affected from birth; described across pediatric and into adult life.

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10. Diagnostics

Genetic testing (diagnostic mainstay). - **Chromosomal microarray (CMA)** detects 6p22-p24 deletions/duplications (first-line for the CNV presentation; DECIPHER/ClinVar). - **Exome (WES) / genome (WGS) sequencing** detects intragenic SNVs/indels and is the primary route to single-gene diagnosis; **GeneMatcher** enabled cohort assembly (Verberne 2021). - **Exon-level copy-number analysis is essential:** intragenic JARID2 CNVs are highly prevalent and may be missed by standard CMA or SNV-only pipelines (Cadieux-Dion 2022, P 35533077: *"the high prevalence of intragenic JARID2 copy number variants, emphasizing the importance of exon-level analysis"*). - **Single-gene / NDD gene-panel** testing including JARID2. - **Karyotype/FISH** only for large/visible rearrangements; **mtDNA and repeat-expansion testing not indicated.**

Omics / epigenomic diagnostics. DNA-methylation episignature (EpiSign) is an emerging clinically validated adjunct that can confirm diagnosis and reclassify VUS in chromatinopathies (Koparir 2026, P 41957673: *"methylation profiles supported the pathogenicity of several variants previously classified as VUS"*); JARID2 genomic boundaries/episignature refinement reported by van der Laan 2023 (P 37762546).

RNA-seq can demonstrate NMD/altered PRC2-target expression (Viitasalo 2022).

Clinical tests / imaging. No specific biomarker or laboratory abnormality. Brain MRI and echocardiography are used to evaluate variable structural anomalies but are **non-diagnostic**. Developmental/neuropsychological assessment quantifies ID/ASD severity.

Clinical criteria & differential diagnosis. No formal consensus diagnostic criteria; diagnosis is **molecular**. Differential diagnosis = other syndromic ID/chromatinopathies (e.g., KMT2A/Wiedemann-Steiner, SETD5-related, ANKRD11/KBG, CHD8, EZH2/Weaver) — distinguished by gene-specific variants, facial gestalt, and episignature. Larger 6p22-p24 deletions must be parsed for ATXN1/DTNBP1 contributions.

Screening. No newborn or population carrier screening (ultra-rare, de novo). **Cascade testing** of parents for a proband's variant informs recurrence risk.

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11. Outcome / Prognosis

- **Survival/mortality:** the dominant NDD is **not life-limiting** per se; normal or near-normal life expectancy expected absent severe congenital organ defects. No disease-specific mortality data (**not available**).
- **Morbidity/disability:** lifelong intellectual disability, speech/communication impairment, and behavioral comorbidity are the principal morbidities; variable motor and adaptive impairment.
- **Quality of life:** no disease-specific QoL instrument data (**not available**).
- **Complications:** behavioral/psychiatric comorbidity; in individuals with larger deletions, congenital organ anomalies (cardiac) and associated complications.
- **Recovery potential:** developmental gains with early intervention; disability is chronic (no cure).
- **Prognostic factors:** variant type/size (contiguous deletions with additional genes tend to be more severe), presence of structural organ anomalies, baseline cognitive/adaptive function. No validated molecular prognostic biomarker.

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12. Treatment

No disease-specific or disease-modifying therapy exists. Management is **symptomatic, supportive, and multidisciplinary**, as for other syndromic NDDs.

- **Pharmacotherapy (symptomatic):** none targeted at JARID2. Standard agents for comorbidities — e.g., stimulants/alpha-agonists for ADHD, SSRIs for anxiety, antiseizure medication if epilepsy present, antipsychotics for severe behavioral dysregulation. **Pharmacogenomics:** no JARID2-specific guidance.
- NCIT examples: Supportive Care (NCIT:C15277); Symptomatic Treatment (NCIT:C15632).
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno): none available / experimental only.** Haploinsufficiency is in principle a target for future dosage-restoring approaches (ASO upregulation, gene therapy), but **no clinical programs exist** for JARID2. PRC2/EZH2 modulators exist in oncology but are **not applicable** therapeutically here.
- **Surgical/interventional:** only for specific congenital anomalies (e.g., cardiac surgery) when present.
- **Supportive & rehabilitative (core management):** early intervention; **physical therapy** (NCIT:C15368), **occupational therapy** (NCIT:C15317), **speech-language therapy** (NCIT:C15312); special education; behavioral therapy (ABA); nutritional support for feeding/hypotonia.
- **Experimental/clinical trials: none registered specifically for JARID2-NDD** (ClinicalTrials.gov — none found).
- **Treatment strategy:** individualized, guided by developmental assessment and comorbidity profile; no genotype-guided algorithm established.

13. Prevention

- **Primary prevention:** not possible (de novo monogenic). **Genetic counseling** is the principal preventive tool.
- **Secondary prevention:** early developmental screening/surveillance enabling early intervention to optimize function.
- **Tertiary prevention:** management of behavioral comorbidity and surveillance for associated anomalies to prevent secondary complications.

- **Reproductive/genetic prevention:** for families with an identified proband variant, **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** are available; low recurrence risk for de novo cases (counsel for possible germline mosaicism). The recessive reproductive phenotype warrants partner carrier consideration in consanguineous families.
- **Immunization / public-health / environmental interventions: not applicable.**

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14. Other Species / Natural Disease

- **Taxonomy:** gene conserved across vertebrates — *Homo sapiens* (NCBI:txid9606); *Mus musculus* (NCBI:txid10090); *Danio rerio* (NCBI:txid7955); *Drosophila melanogaster* (NCBI:txid7227, ortholog *jumonji/Jarid2/dJmj*).
- **Orthologous genes:** mouse **Jarid2** (NCBI Gene 16468; historical name *jmj*); zebrafish **jarid2a/jarid2b**; *Drosophila* **Jarid2/dJmj**.
- **Natural disease in other species: no naturally occurring JARID2 disease reported** in companion animals/wildlife (OMIA — none recorded). Phenotypes are experimentally induced.
- **Comparative biology:** the PRC2/H3K27me3 regulatory mechanism is **deeply evolutionarily conserved**; mouse jumonji mutants model the developmental requirement (neurulation, cardiogenesis). *Drosophila* dJmj has partly PRC2-independent roles (Suong 2018,

  P 29102770

  ), indicating some species divergence in function.
- **Transmission / zoonosis: not applicable** (genetic disorder).

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15. Model Organisms

- **Mouse (primary model):** gene-trap/knockout **jumonji (jmj/Jarid2)** mice show fully penetrant **neural tube defects and cardiac outflow-tract/trabecular abnormalities** on a C3H/HeJ background, with **strain-dependent penetrance** (Takeuchi 1999,

:

P 10446263

"the jmj gene plays essential roles in the normal development of the neural plate, morphogenesis of bulbus cordis, and proliferation of trabecular myocytes"). Available model types: constitutive knockout, gene-trap, conditional alleles (MGI). Homozygous null is embryonic/perinatal lethal, so conditional/heterozygous models better mimic human haploinsufficiency.

- **In-vitro / cellular: mouse embryonic stem cells (ESCs)** are the workhorse for PRC2.2/JARID2 recruitment and H3K27me3 studies (Loh & Veenstra 2022; Granat 2026; Agius 2026). Reconstituted PRC2 biochemistry defines JARID2's H3K27me3-mimic function. Mouse zygote mRNA-microinjection models the recessive p.Arg300Gln allele (Sun 2025).
- **Invertebrate:** *Drosophila* dJmj (Suong 2018) for stem-cell/differentiation roles (partly PRC2-independent).
- **Phenotype recapitulation:** mouse captures the **developmental CNS and cardiac requirement** and the PRC2/H3K27me3 mechanism well; it does **not** directly model human cognition/behavior or the adult neurodevelopmental phenotype. iPSC-derived neurons/organoids from patients are a logical but **not-yet-prominently-published** resource (**gap**).
- **Resources:** MGI (mouse), ZFIN (zebrafish), FlyBase (Drosophila), IMPC/IMSR for alleles.

Evidence Summary (key PMIDs)

PMID	Type	Contribution
33077894 (Verberne 2021)	Human clinical cohort (n=16)	Establishes JARID2 haploinsufficiency as a distinct NDD; gene–disease validity
35533077 (Cadieux-Dion 2022)	Human clinical (n=7/5 families)	Confirms gene–disease assoc.; adds palatal/cardiac anomalies; variable expressivity; intragenic CNVs
23294540 (Barøy 2013)	Human clinical (6p22.3 deletions)	Defines facial gestalt; implicates JARID2 + ATXN1
35979655 (Viitasalo 2022)	Human case + RNA-seq	Duplication (gain) also pathogenic; PRC2-target dysregulation
40886009 (Sun 2025)	Human + mouse zygote	Recessive missense; ERK1/2 + H3K27me3 link; maternal-effect
10446263 (Takeuchi 1999)	Mouse model	Neural-tube + cardiac developmental requirement
35997369 (Loh 2022)	Review/in vitro	JARID2 recruits PRC2 (PRC2.2) in ESCs
41785858 (Agius 2026)	Mouse/in vitro	JARID2 mimics H3K27me3 to seed PRC2 read-write loop
41650228 (Granat 2026)	In vitro	MTF2/JARID2 direct PRC2 to chromatin; H3K27me3 deposition
41957673 (Koparir 2026)	Human diagnostic	Episignature/EpiSign for chromatinopathy VUS resolution
37762546 (van der Laan 2023)	Human diagnostic	Refines JARID2 genomic boundaries/episignature

Limitations and Future Directions

- The disorder is **newly defined with only a few dozen reported individuals**; frequencies, prevalence, sex ratio, penetrance, and prognosis are **preliminary** and from small cohorts.
- **Phenotype frequencies are qualitative**; no large natural-history study exists.
- Human **neuronal/organoid models and a fully validated JARID2 episignature classifier** are still maturing.
- The relationship between **dosage loss vs gain**, and the **ERK1/2 branch**, needs mechanistic clarification in neural tissue.
- **No disease-specific therapy**; dosage-restoration strategies are hypothetical.

Ontology term appendix (suggested)

- **Disease:** MONDO:0859306
- **Gene/protein:** HGNC:6826 (JARID2); UniProt Q92833
- **GO BP:** GO:0070734 (histone H3-K27 methylation), GO:0006342 (chromatin silencing), GO:0021915 (neural tube development), GO:0030182 (neuron differentiation), GO:0045596 (neg. reg. of cell differentiation)
- **GO CC:** GO:0005634 (nucleus), GO:0000785 (chromatin), GO:0035098 (ESC/ATAC-type complex context: PRC2 complex GO:0035098)
- **CL:** CL:0000047 (neural stem cell), CL:0000540 (neuron), CL:0000746 (cardiomyocyte)
- **UBERON:** UBERON:0000955 (brain), UBERON:0001016 (nervous system), UBERON:0000948 (heart), UBERON:0005732 (neural plate)
- **HPO:** HP:0001263, HP:0001249, HP:0000750, HP:0001252, HP:0000708, HP:0000729, HP:0000336, HP:0000490, HP:0011800
- **NCIT (interventions):** NCIT:C15368 (physical therapy), NCIT:C15317 (occupational therapy), NCIT:C15312 (speech therapy), NCIT:C15277 (supportive care)
